

```
import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for mathematical matrix o
import matplotlib.pyplot as plt # Import Matplotlib to visualize the out

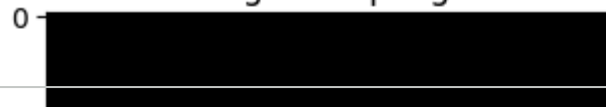
# Step 1: Create an image with a clear, readable letter 'A'
img = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array filled with zeros
cv2.putText(img, 'A', (20, 80), cv2.FONT_HERSHEY_SIMPLEX, 3, 255, 5) # Render text characters onto t

# Step 2: Simulate an extremely low sampling rate (sensor with few pixels)
low_res = cv2.resize(img, (10, 10)) # Resize image to new dimensions using interpolation

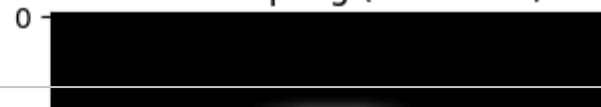
# Step 3: Attempt to resize it back up using interpolation to 'enhance' it
# This proves that lost sampling data cannot simply be interpolated back
upscaled = cv2.resize(low_res, (100, 100), interpolation=cv2.INTER_LINEAR) # Resize image to new dim

# Step 4: Show the permanent loss of visibility
plt.figure(figsize=(8,4)) # Create a new figure window for display
plt.subplot(121), plt.imshow(img, cmap='gray'), plt.title('High Sampling') # Place this image in a s
plt.subplot(122), plt.imshow(upscaled, cmap='gray'), plt.title('Low Sampling (Data Lost)') # Place t
plt.show() # Show the final plot window to the user
```

High Sampling



Low Sampling (Data Lost)



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